Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: Batch – 13

Start date: 02 February 2026

S.NO	Team Members Name
1.	Sridhar Y Gowdar
2.	Saghana S
3.	Akshita
4.	Malavika K C
5.	Sneha Kokonda
6.	Kavipriya S
7.	Abinaya Durai
8.	Ashish
9.	Poorna Prakash
10.	Guntupalli Venkat Sreenadh
11.	Akhil Nagavelli
12.	B Tharshith Vardhan
13.	Kaluva Praneeth Kumar
14.	Gangatharan M
15.	Pavan Powar
16.	Nivetha S
17.	Vigneshwaran K
18.	Mayur Ukandji Narwade

Internship Duration: 8 Weeks

1. Project Title

Web Application for Smart e-Waste Collection and Management.

2. Project Objective

The main objective of this project was to develop a digital platform to manage electronic waste efficiently by connecting users, recyclers, and collection centers. It aims to reduce improper disposal, promote recycling, and support environmental sustainability.

This project aims to promote responsible e-waste disposal, improve operational efficiency, and provide a user-friendly solution for both users and administrators.

3. Project description in detail

The Smart e-Waste Collection and Management System is designed to provide a complete digital solution for handling electronic waste collection.

Approach

The project was developed using a modular approach, where each functionality (authentication, request submission, admin management, collector verification, and notifications) was implemented step-by-step. Agile methodology was followed to ensure continuous development and testing.

Technologies Used

- **Frontend:** React (JavaScript, HTML, CSS)
- **Backend:** Spring Boot (Java)
- **Database:** MySQL
- **Email Service:** JavaMail API and SMTP
- **API Used:** Mapbox API (for location selection), Cloudinary (for image storage)
- **Tools:** Git, Postman

Key Features

- User registration and login system
- E-waste request submission with detailed inputs
- Admin dashboard for managing requests
- Pickup scheduling system
- Collector dashboard for managing assigned pickup requests and confirming collection
- OTP-based pickup verification for ensuring actual completion of requests
- Automated email notifications
- Real-time status tracking

Real-World Impact

This system helps reduce improper disposal of electronic waste by providing an easy and structured method for collection. It improves environmental sustainability and ensures safe handling of e-waste.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project planning, requirement analysis	Completed requirement analysis and design
Week 2	User authentication module	Implemented login and registration
Week 3	E-waste submission module	Developed request form and database integration
Week 4	Request tracking feature	Enabled status tracking
Week 5	Admin dashboard development	Built request management system
Week 6	Scheduling & email notifications	Implemented scheduling and email alerts
Week 7	Testing and debugging	Fixed bugs and improved performance
Week 8	Documentation and final review	Prepared final report and submission

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial planning and requirement gathering	Milestone 1 15-02-2026
Prototype/First Draft	Basic working model with core features	Milestone 2 01-03-2026
Mid-Term Review	Evaluation of system progress	Milestone 3 15-03-2026
Final Submission	Completion of full system	Milestone 4 22-03-2026

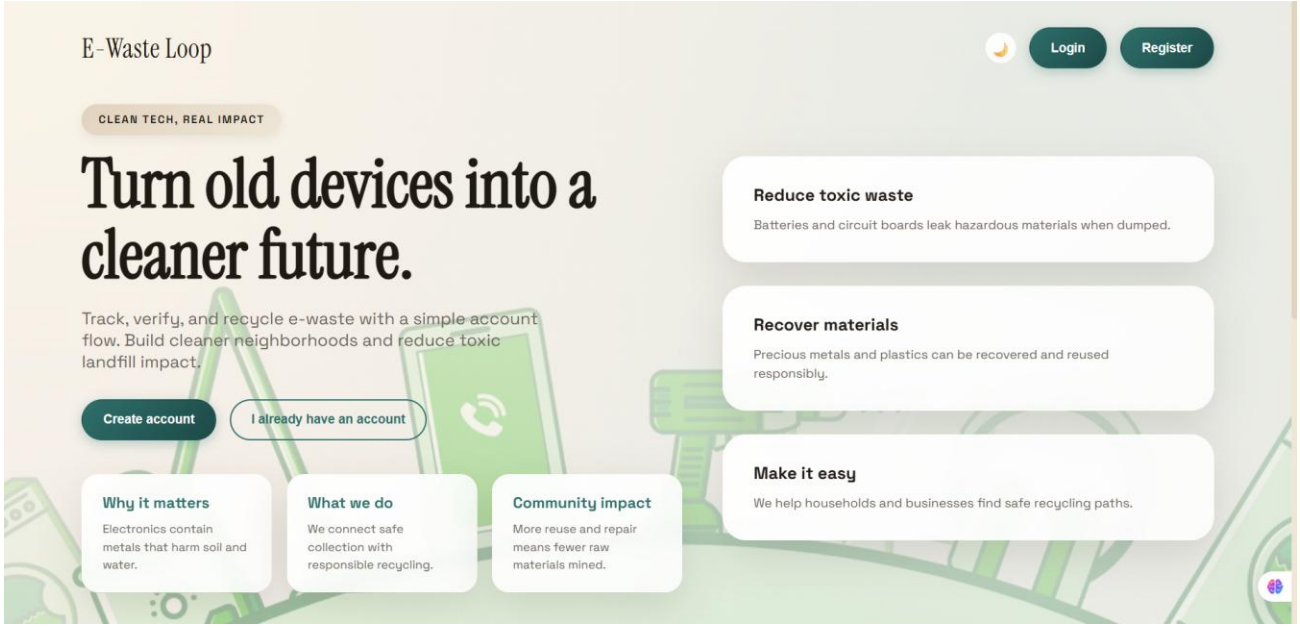
5b. Project execution details

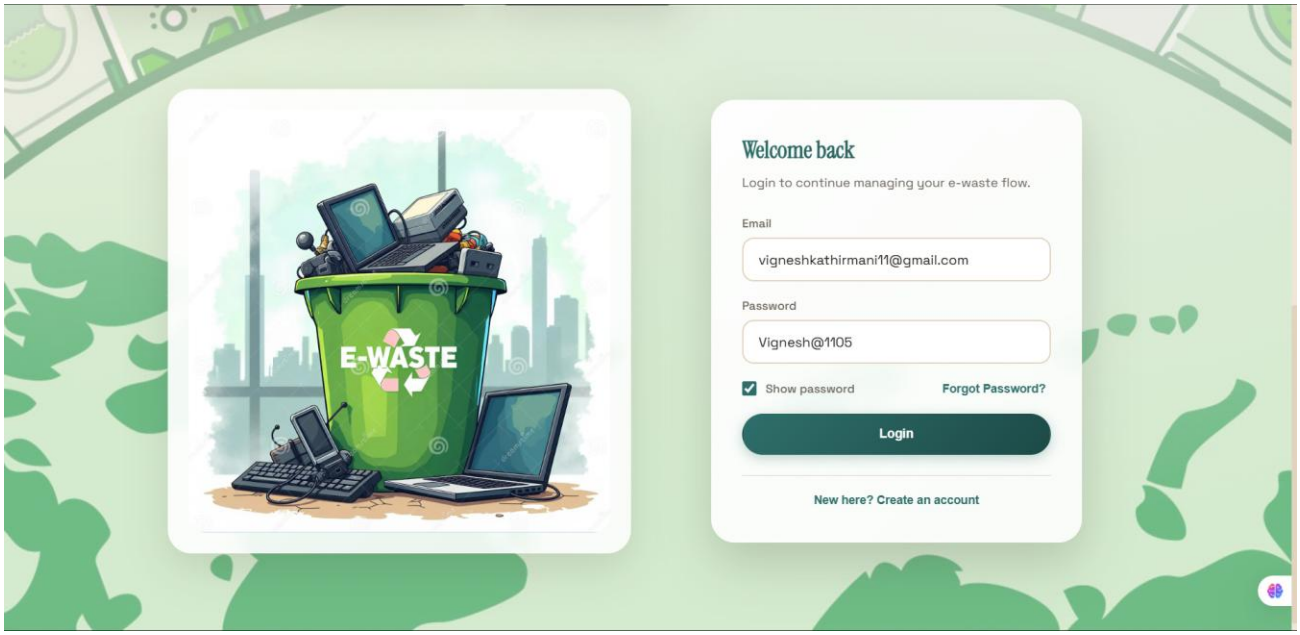
The project was executed in a structured and phased manner:

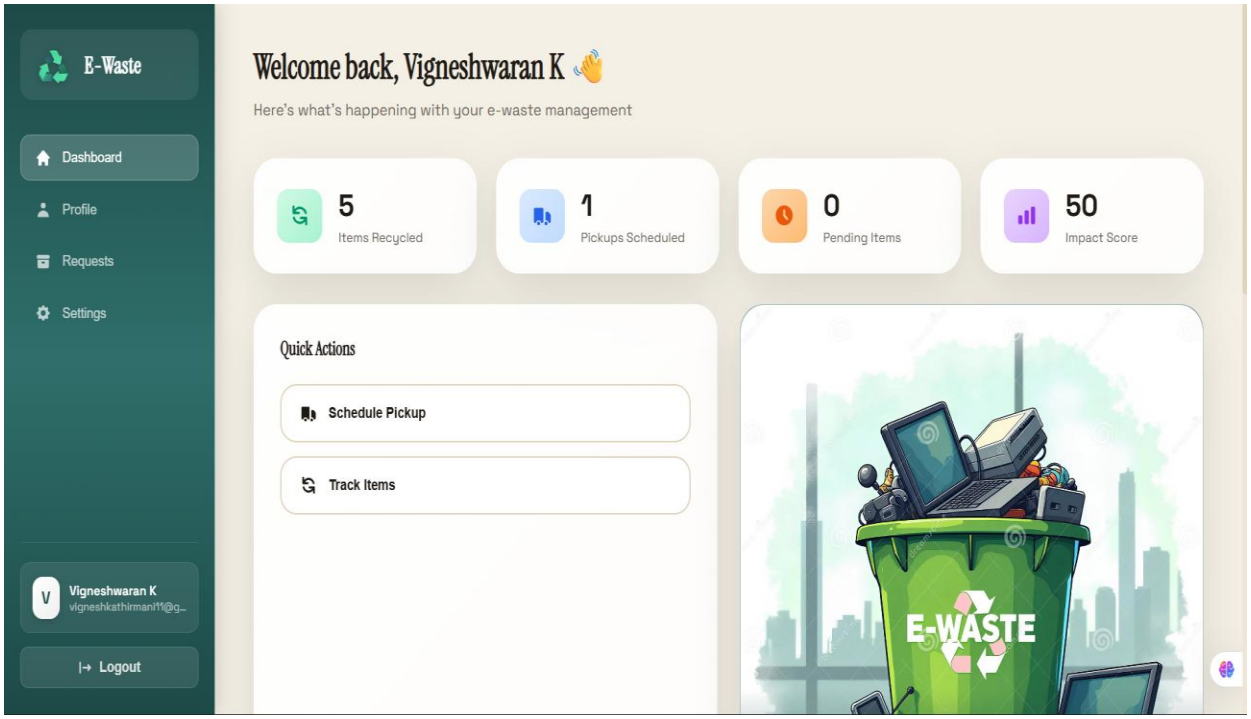
- Initially, requirements were gathered and system design was planned.
- The user authentication module was developed to ensure secure access using OTP-based verification.
- The e-waste request module was implemented with proper input validation and database storage.
- The admin dashboard was developed to manage requests, approve or reject submissions, and schedule pickups efficiently.
- A collector (delivery person) module was introduced, where the admin assigns approved requests to specific collectors based on location.
- The collector dashboard was developed to allow collectors to view assigned requests and perform pickup operations.
- To ensure real-world reliability, an OTP-based pickup verification mechanism was implemented. The collector verifies the pickup using an OTP provided by the user, and only after successful verification, the request is marked as completed.
- Email notification functionality was integrated to provide updates to users regarding OTP verification and request status.
- Mapbox API was integrated for accurate location selection during request submission.
- Testing was conducted at multiple levels (unit, integration, and system testing).
- Bugs were fixed and performance was optimized before final deployment.

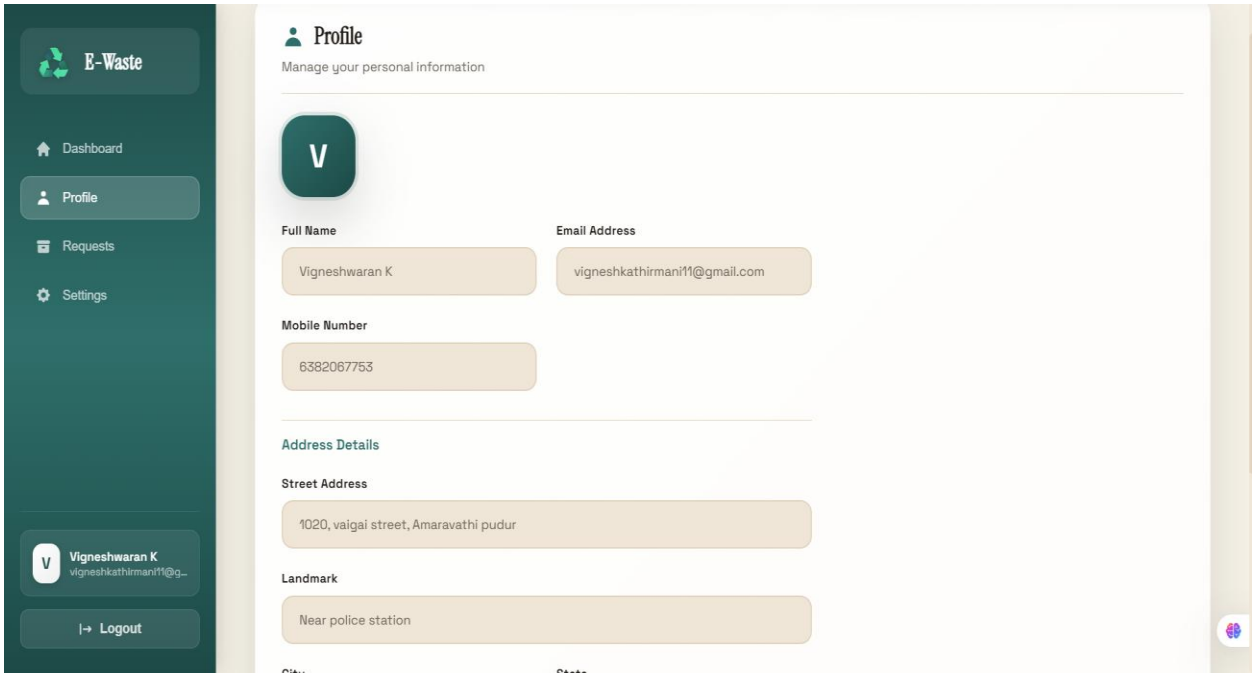
The project followed a systematic workflow ensuring each module was properly integrated and tested.

6. Snapshots / Screenshots






E-Waste
Dashboard
Profile
Requests
Settings
Vigneshwaran K
vigneshkathirmani11@g...
Logout
Welcome back, Vigneshwaran K
Here's what's happening with your e-waste management
5
Items Recycled
1
Pickups Scheduled
0
Pending Items
50
Impact Score
Quick Actions
Schedule Pickup
Track Items
E-WASTE


E-Waste
Dashboard
Profile
Requests
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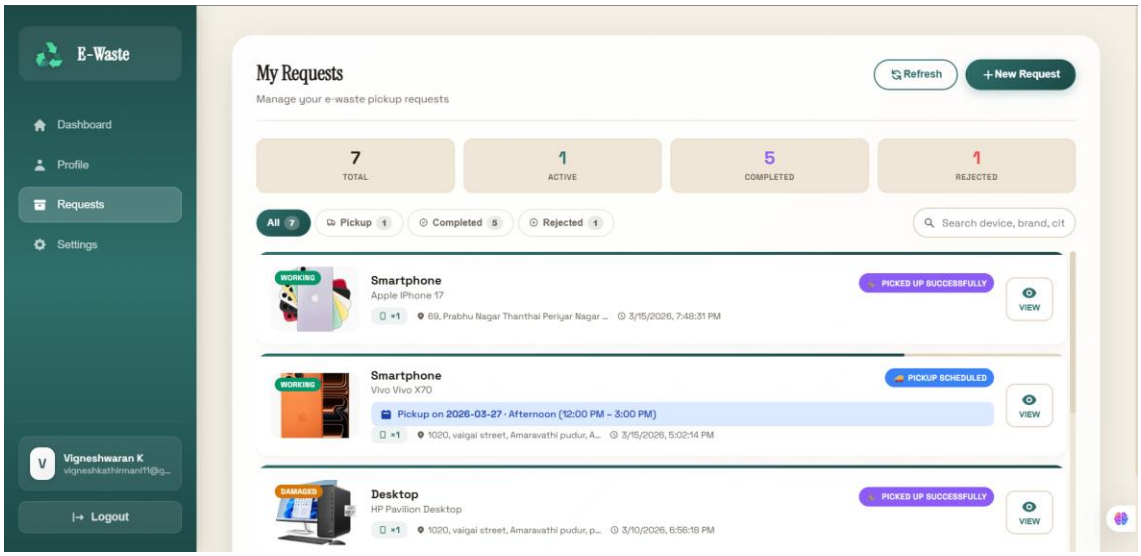
vigneshkathirmani11@g...
Logout
Profile
Manage your personal information
V
Full Name
Vigneshwaran K
Email Address
vigneshkathirmani11@gmail.com
Mobile Number
6382067753
Address Details
Street Address
1020, vaigai street, Amaravathi pudur
Landmark
Near police station


E-Waste
Dashboard
Profile
Requests
Settings
Vigneshwaran K
vigneshkathirmani11@g...
Logout
My Requests
Manage your e-waste pickup requests
Refresh
+ New Request
7
TOTAL
1
ACTIVE
5
COMPLETED
1
REJECTED
All 7
Pickup 1
Completed 5
Rejected 1
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Smartphone
Apple IPhone 17
×1
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3/15/2026, 7:48:31 PM
PICKED UP SUCCESSFULLY
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Smartphone
Vivo Vivo X70
PICKUP SCHEDULED
Pickup on 2026-03-27 · Afternoon (12:00 PM – 3:00 PM)
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3/15/2026, 5:02:14 PM
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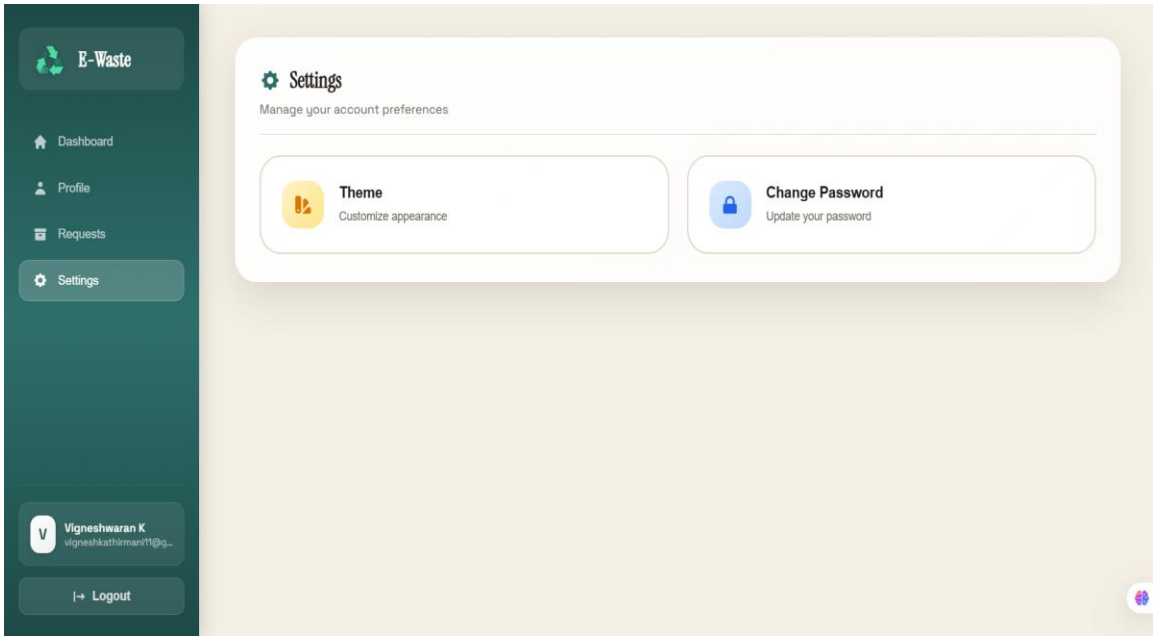
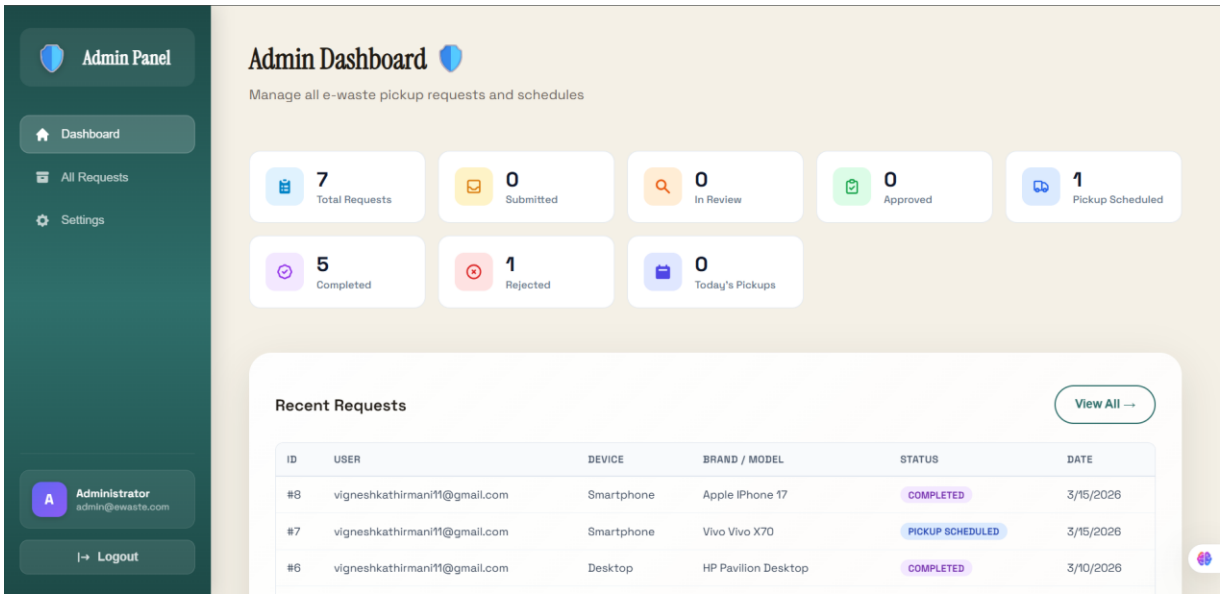
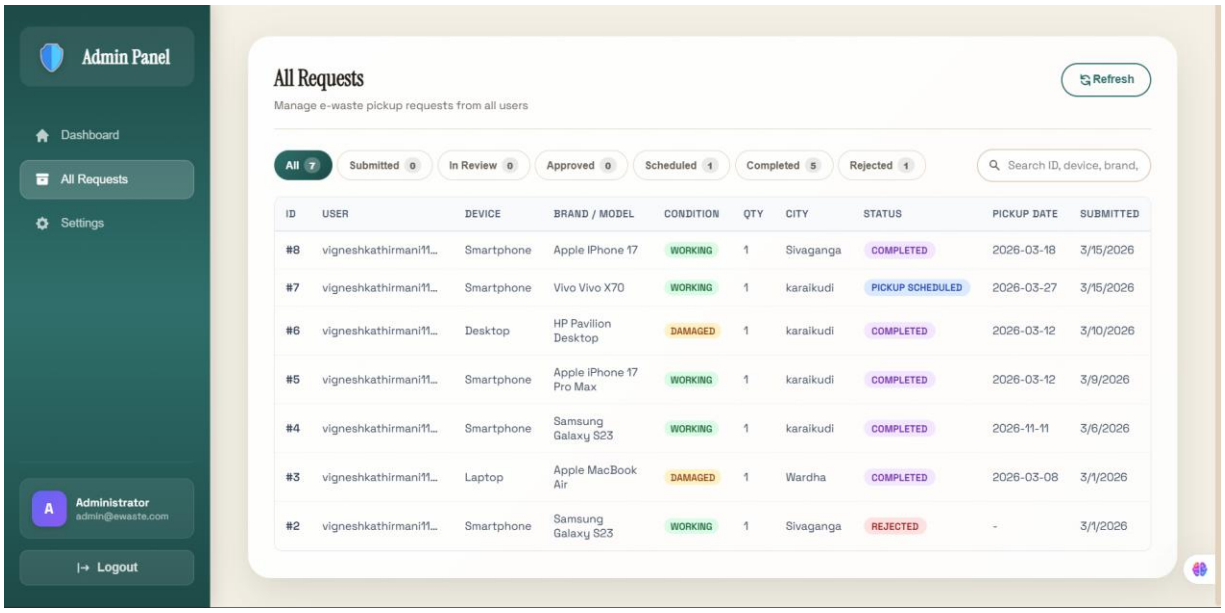
PICKED UP SUCCESSFULLY
VIEW


E-Waste
Dashboard
Profile
Requests
Settings
Vigneshwaran K
vigneshkathirmani11@g...
Logout
Settings
Manage your account preferences
Theme
Customize appearance
Change Password
Update your password


Admin Panel
Dashboard
All Requests
Settings
Administrator
admin@ewaste.com
Logout
Admin Dashboard
Manage all e-waste pickup requests and schedules
7 Total Requests
0 Submitted
0 In Review
0 Approved
1 Pickup Scheduled
5 Completed
1 Rejected
0 Today's Pickups
Recent Requests
View All →
ID USER DEVICE BRAND / MODEL STATUS DATE
#8 vigneshkathirmani11@gmail.com Smartphone Apple IPhone 17 COMPLETED 3/15/2026
#7 vigneshkathirmani11@gmail.com Smartphone Vivo Vivo X70 PICKUP SCHEDULED 3/15/2026
#6 vigneshkathirmani11@gmail.com Desktop HP Pavilion Desktop COMPLETED 3/10/2026


Admin Panel
Dashboard
All Requests
Settings
Administrator
admin@ewaste.com
Logout
All Requests
Manage e-waste pickup requests from all users
Refresh
All 7 Submitted 0 In Review 0 Approved 0 Scheduled 1 Completed 5 Rejected 1
Search ID, device, brand,
ID USER DEVICE BRAND / MODEL CONDITION QTY CITY STATUS PICKUP DATE SUBMITTED
#8 vigneshkathirmani11... Smartphone Apple IPhone 17 WORKING 1 Sivaganga COMPLETED 2026-03-18 3/15/2026
#7 vigneshkathirmani11... Smartphone Vivo Vivo X70 WORKING 1 karaikudi PICKUP SCHEDULED 2026-03-27 3/15/2026
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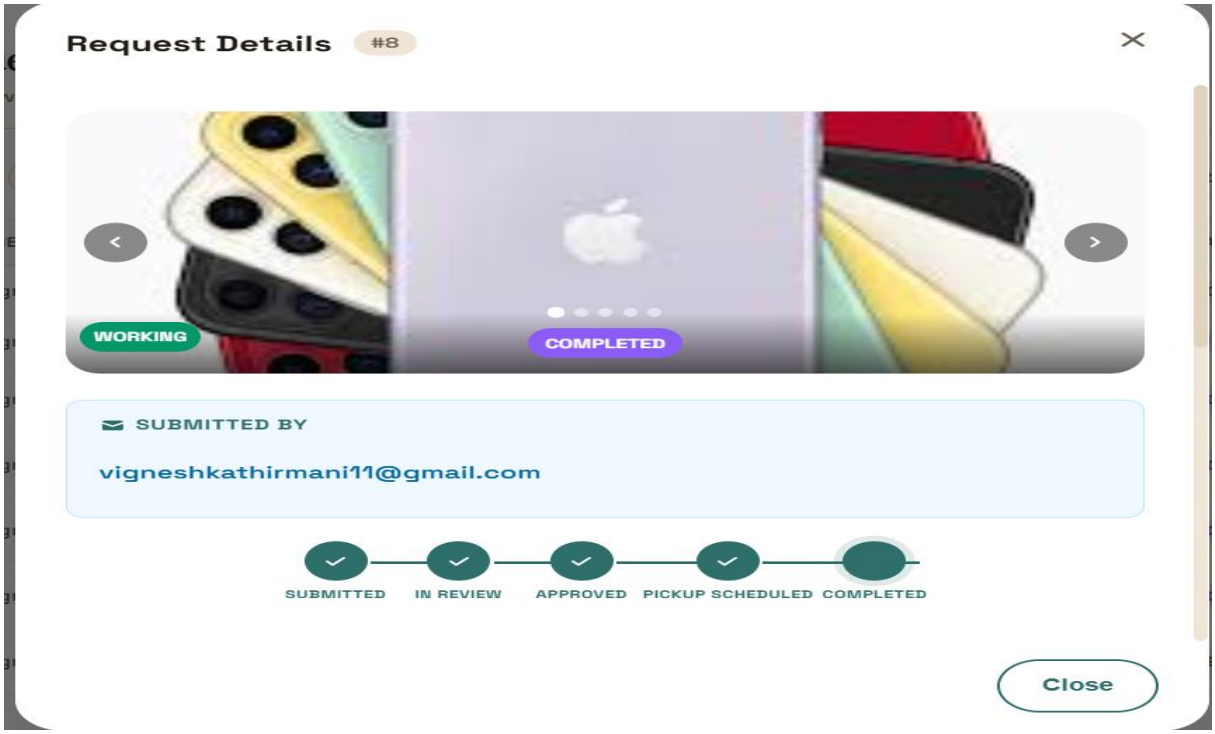
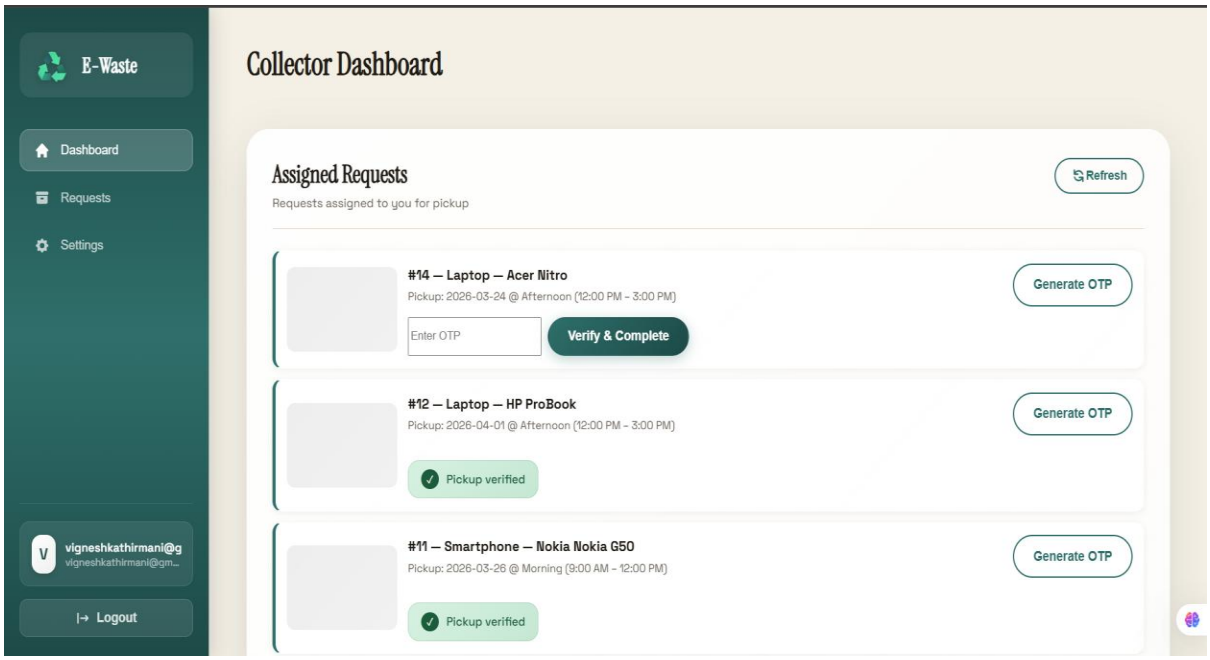
#5 vigneshkathirmani11... Smartphone Apple iPhone 17 Pro Max WORKING 1 karaikudi COMPLETED 2026-03-12 3/9/2026
#4 vigneshkathirmani11... Smartphone Samsung Galaxy S23 WORKING 1 karaikudi COMPLETED 2026-11-11 3/6/2026
#3 vigneshkathirmani11... Laptop Apple MacBook Air DAMAGED 1 Wardha COMPLETED 2026-03-08 3/1/2026
#2 vigneshkathirmani11... Smartphone Samsung Galaxy S23 WORKING 1 Sivaganga REJECTED - 3/1/2026


Request Details
#8
WORKING
COMPLETED
SUBMITTED BY
vigneshkathirmani11@gmail.com
SUBMITTED
IN REVIEW
APPROVED
PICKUP SCHEDULED
COMPLETED
Close


E-Waste
Dashboard
Requests
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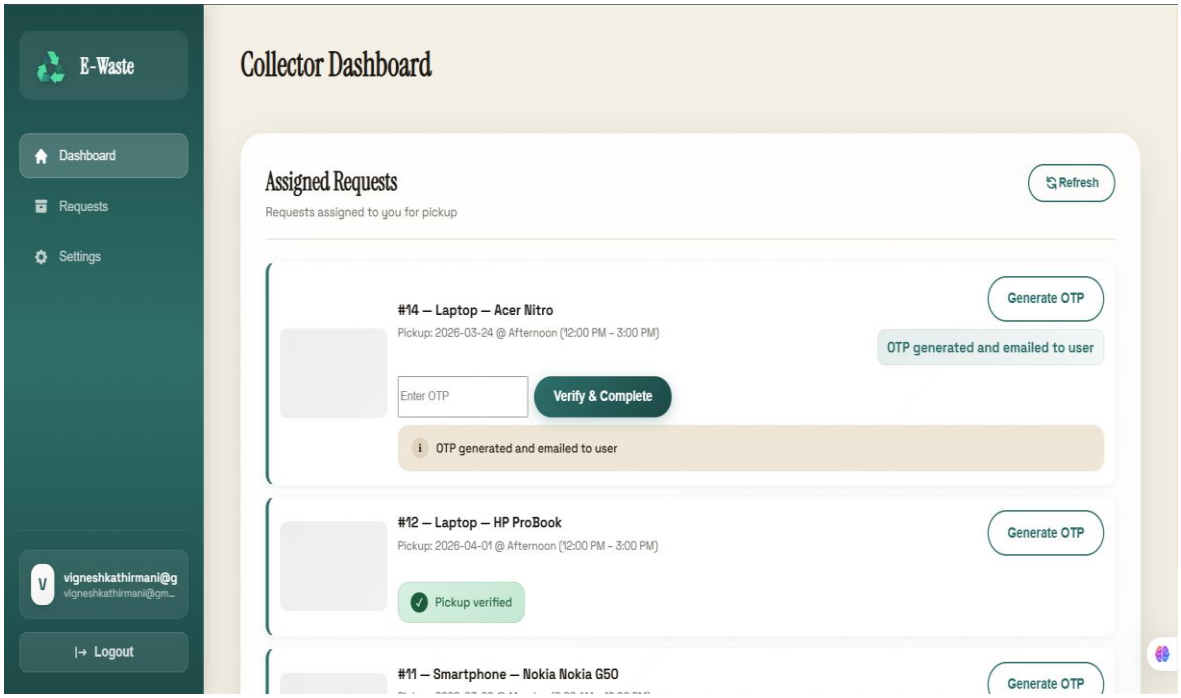
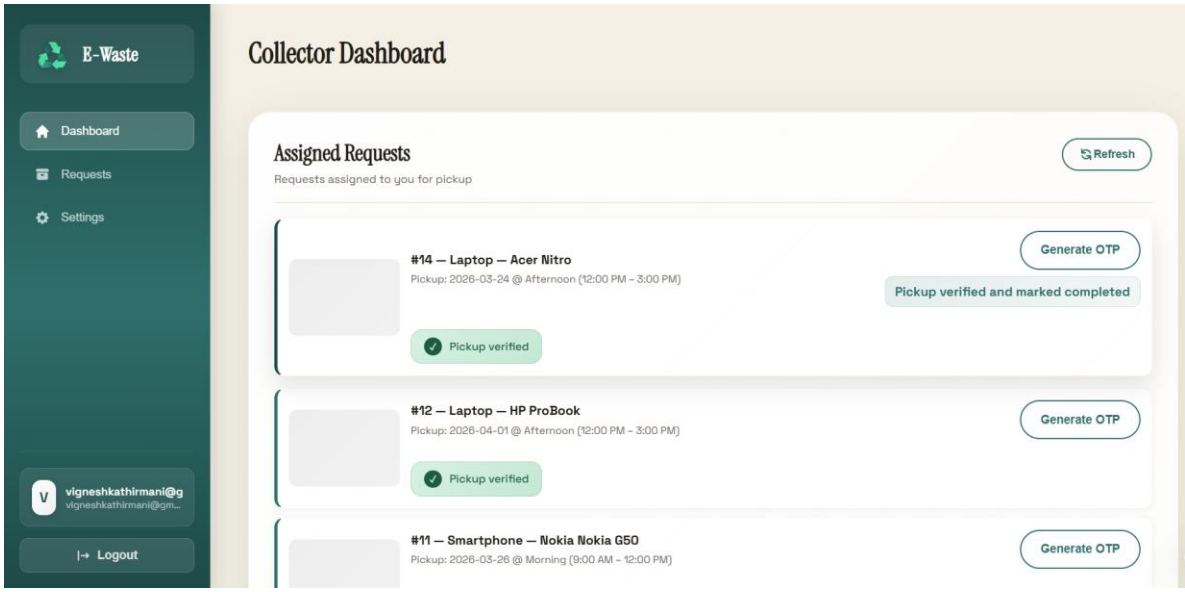
Logout
Collector Dashboard
Assigned Requests
Requests assigned to you for pickup
Refresh
#14 — Laptop — Acer Nitro
Pickup: 2026-03-24 @ Afternoon (12:00 PM - 3:00 PM)
Enter OTP
Verify & Complete
Generate OTP
#12 — Laptop — HP ProBook
Pickup: 2026-04-01 @ Afternoon (12:00 PM - 3:00 PM)
Pickup verified
Generate OTP
#11 — Smartphone — Nokia Nokia G50
Pickup: 2026-03-26 @ Morning (9:00 AM - 12:00 PM)
Pickup verified
Generate OTP


E-Waste
Dashboard
Requests
Settings
vigneshkathirmani@g
vigneshkathirmani@gm...
Logout
Collector Dashboard
Assigned Requests
Requests assigned to you for pickup
Refresh
#14 — Laptop — Acer Nitro
Pickup: 2026-03-24 @ Afternoon (12:00 PM – 3:00 PM)
Generate OTP
OTP generated and emailed to user
Enter OTP
Verify & Complete
OTP generated and emailed to user
#12 — Laptop — HP ProBook
Pickup: 2026-04-01 @ Afternoon (12:00 PM – 3:00 PM)
Generate OTP
Pickup verified
#11 — Smartphone — Nokia Nokia G50
Generate OTP


E-Waste
Dashboard
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vigneshkathirmani@g
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Logout
Collector Dashboard
Assigned Requests
Requests assigned to you for pickup
Refresh
#14 — Laptop — Acer Nitro
Pickup: 2026-03-24 @ Afternoon (12:00 PM – 3:00 PM)
Generate OTP
Pickup verified and marked completed
Pickup verified
#12 — Laptop — HP ProBook
Pickup: 2026-04-01 @ Afternoon (12:00 PM – 3:00 PM)
Generate OTP
Pickup verified
#11 — Smartphone — Nokia Nokia G50
Pickup: 2026-03-26 @ Morning (9:00 AM – 12:00 PM)
Generate OTP

7. Challenges Faced

1. Database Integration Issues

- Initially, there were difficulties in establishing a stable connection between the Spring Boot backend and the MySQL database, which caused issues in storing and retrieving data.
- **Solution:** This was resolved by properly configuring the database connection properties and thoroughly testing the APIs using Postman to ensure smooth data flow.

2. Email Notification Setup

- Configuring automated email notifications for OTP verification and request updates was challenging due to SMTP authentication and configuration issues.
- **Solution:** The issue was resolved by correctly configuring the JavaMail API with appropriate SMTP settings and verifying the email service functionality.

3. Real-Time Status Updates

- **Challenge:** Maintaining synchronization between user and admin views for request status updates was difficult, as changes were not immediately reflected across the system.
- **Solution:** This was addressed by implementing proper REST API calls and refresh mechanisms to ensure consistent and up-to-date data across all interfaces.

4. Debugging Errors

- **Challenge:** During the integration of multiple modules, unexpected bugs and errors occurred, affecting the overall system functionality.
- **Solution:** These issues were resolved through systematic debugging, testing individual components, and ensuring proper integration between modules.

5. Pickup Verification Challenge

- **Challenge:** Initially, the system allowed the admin to mark requests as completed after scheduling without verifying whether the actual pickup had taken place, leading to an assumption-based workflow**.**
- **Solution:** To overcome this, a collector (delivery person) module was introduced along with an OTP-based verification mechanism. The collector verifies the

pickup using an OTP provided by the user, ensuring that the request is marked as completed only after actual confirmation.

8. Learnings & Skills Acquired

Technical Skills

- Backend development using Spring Boot
- Database design and management (MySQL)
- REST API development
- Email integration
- Debugging and testing

Tools Learned

- Git (Version Control)
- Postman (API Testing)

Soft Skills

- Problem-solving
- Time management
- Team collaboration

9. Testimonials from team

During this internship, I had a great learning experience working on a real-world project. The guidance from my mentor helped me understand both technical and practical aspects of software development.

I successfully contributed to building a functional system and improved my confidence in handling full-stack development tasks.

10. Conclusion

The Smart e-Waste Management System was successfully developed and implemented. The project provides an efficient solution for managing e-waste collection through a user-friendly platform. This internship helped me gain practical experience in software development and enhanced my technical and problem-solving skills. It also strengthened my understanding of real-world application development.

11. Acknowledgements

I would like to express my sincere gratitude to my organization and mentor for providing me with this opportunity. Their continuous guidance and support helped me complete this project successfully.

I also thank my team members and peers for their collaboration and valuable suggestions throughout the internship.